Lecture #7

471C / 381V Wireless Communications Lab

From Previous Lecture

- Can use the complex envelope concept to represent the impact of an LTI system on a passband channel entirely at baseband
- Can sample the complex baseband equivalent channel to create the discrete-time equivalent system
- Complex pulse amplitude modulation techniques is a modulation technique where complex symbols ride a succession of pulse shapes
- The complex symbols come from a finite constellation

Preview of Today's Lecture

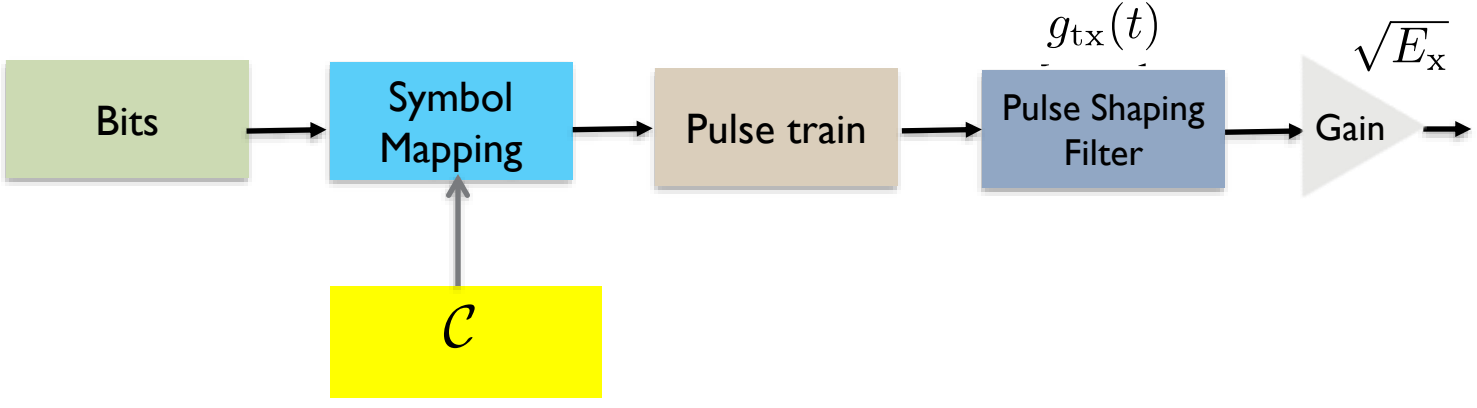
- Constellations
  - Draw a constellation diagram including bit labels (last-time)
  - Calculate the energy of a constellation and normalize it
- Computing the bandwidth, energy, and power of x(t)
  - Calculate bandwidth, energy, and power of a general complex pulse amplitude modulated signal
- Additive white Gaussian noise (AWGN) channel
  - Define AWGN including the distribution and properties
- Overview of AWGN transmitter and receiver
  - Describe key components of the transmitter and receiver
- Pulse shape design
  - Determine properties of the optimal receiver pulse-shape for the AWGN channel

Constellations

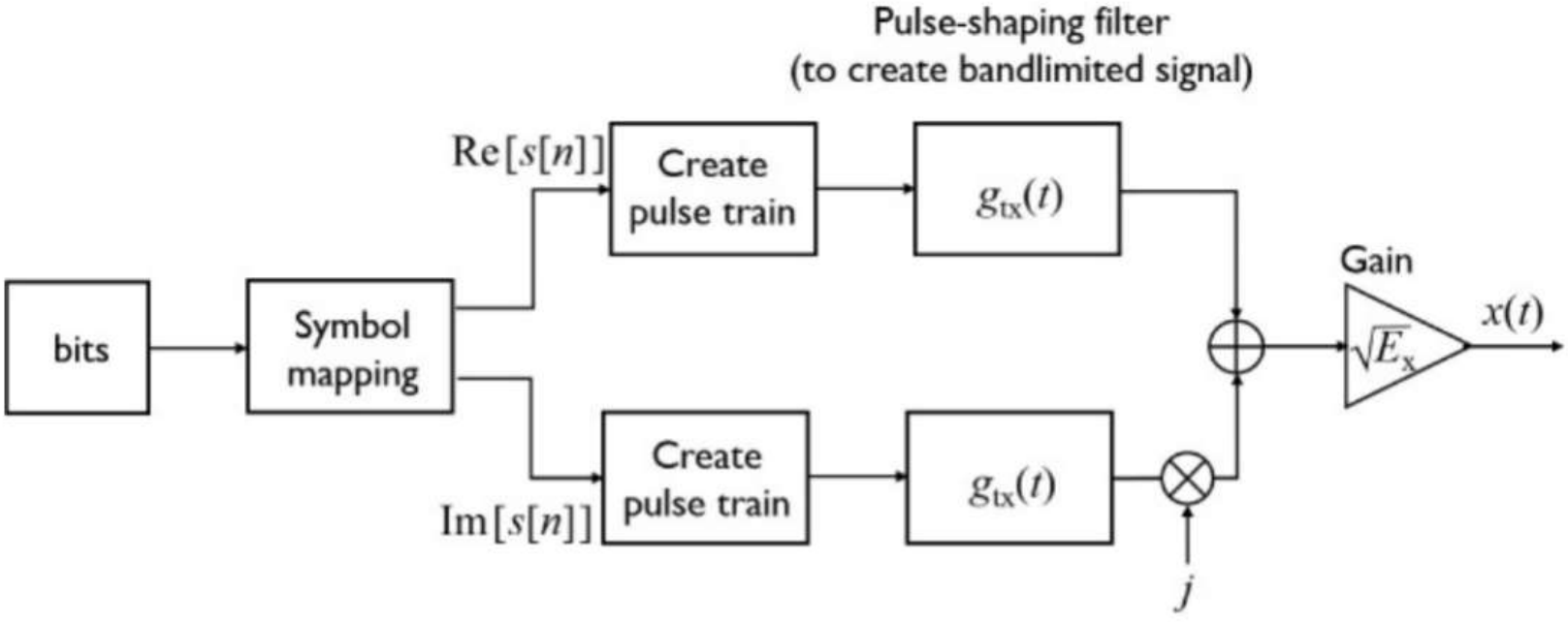
Learning objectives:

- Draw a constellation diagram including bit labels
- Calculate the energy of a constellation and normalize it

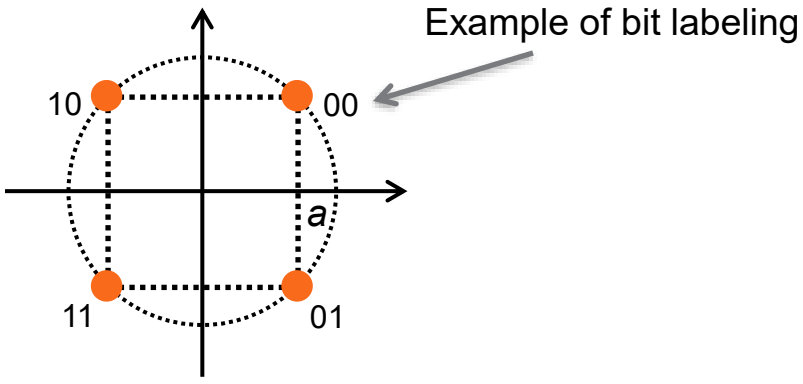
Complex pulse amplitude modulation



- Focus in this part on the constellation


Pulse-shaping filter
(to create bandlimited signal)
Re[s[n]]
Create pulse train
$g_{tx}(t)$
bits
Symbol mapping
Gain
$\sqrt{E_x}$
$x(t)$
Create pulse train
$g_{tx}(t)$
Im[s[n]]
j

Constellations



- A constellation is defined by
  - A set of symbols or alphabet
  - A bit-to-symbol mapping
- Bit labeling is typically done according to the Gray coding principle
  - Adjacent symbols differ by a single binary digit
- Assumptions made about constellations used in this course
  - They are equally likely
  - They are zero-mean
  - They are unit energy

Common constellations (1)

- Binary Phase Shift Keying (BPSK)

$$\mathcal{C} = \{+1, -1\}$$

- M-PAM is a generalization of BPSK to any M that is a power of 2

$$\mathcal{C} = \left\{ -\frac{(M-1)}{2}, \ldots, -\frac{1}{2}, \frac{1}{2}, \ldots, \frac{(M-1)}{2} \right\}$$

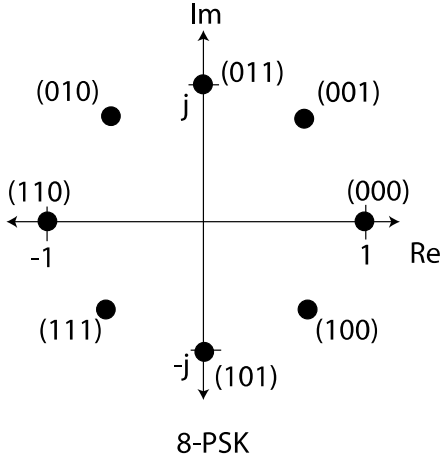
- Quadrature phase shift keying (QPSK) is a complex generalization of BPSK

$$\mathcal{C} = \{1 + \mathrm{j}, -1 + \mathrm{j}, -1 - \mathrm{j}, 1 - \mathrm{j}, \}$$

Common constellations (2)

- M-Phase Shift Keying (M-PSK)

$$\mathcal{C} = \left\{ e^{\frac{\mathrm{j}2\pi k}{M}} \right\}_{k=0}^{M-1}$$

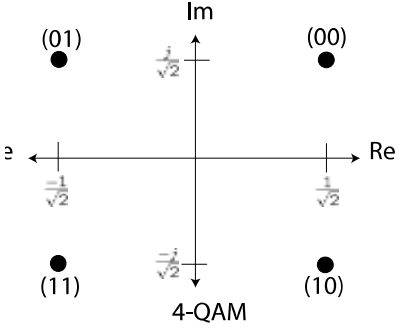


Common constellations (3)

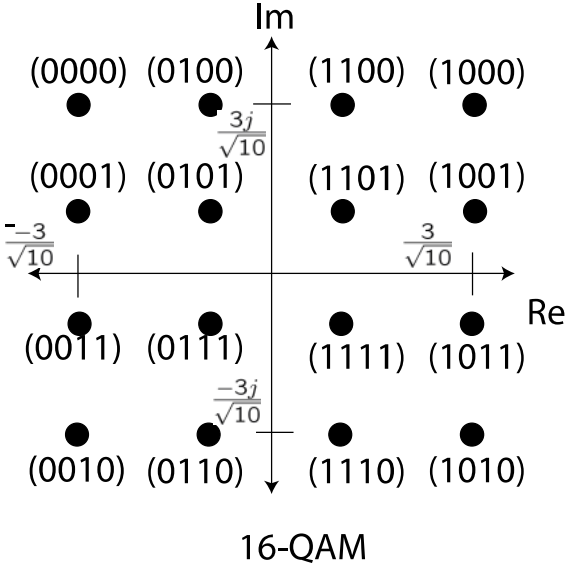
- M-QAM, generalization of QPSK to arbitrary M, power of 4

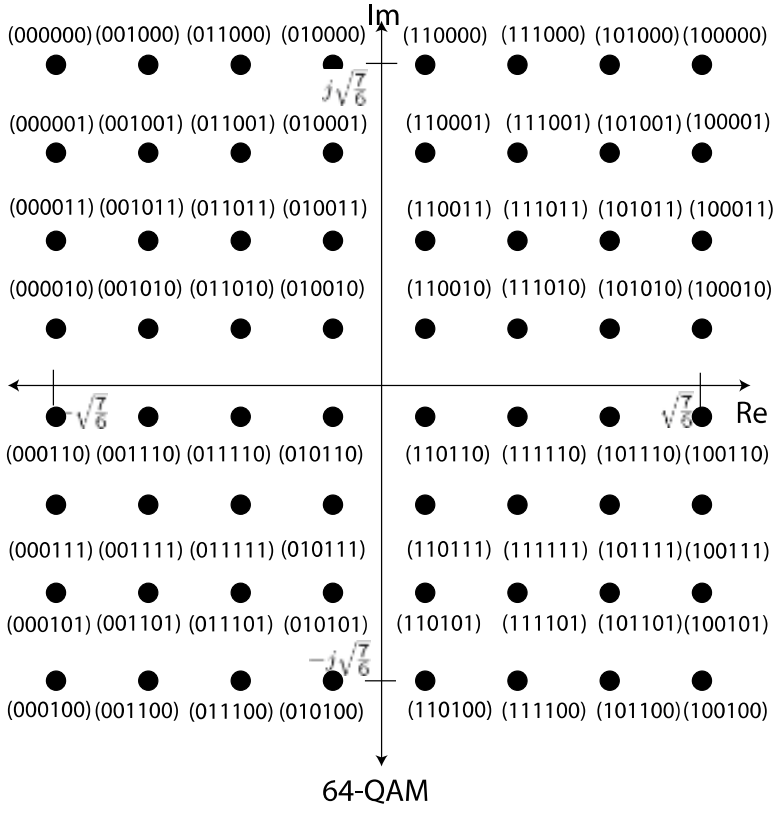
$$\mathcal{C} = \left\{ \cdots, \frac{k}{2} + \mathrm{j}\frac{\ell}{2}, \cdots \right\} \quad k, \ell \in \{-(M/2-1), -(M/2-2), \ldots, -1, 1, \ldots, (M/2-1)\}$$

  - 4, 16, and 64 QAM are common in cellular and WLAN
  - 256 QAM is used in 802.11ac
  - Higher order QAM is used in microwave backhaul



Common constellations (4)





Mean of a constellation

$$\mu_s = \mathbb{E}_s\left[s[n]\right]$$

- Because the symbols are equally likely the mean is

$$\mu_s = \frac{1}{M}\sum_{m=0}^{M-1} s_m$$

- Most wireless constellations are zero-mean
  - Otherwise there would be a DC component
  - DC component is problematic at baseband

Constellation energy

- The constellation energy is defined as

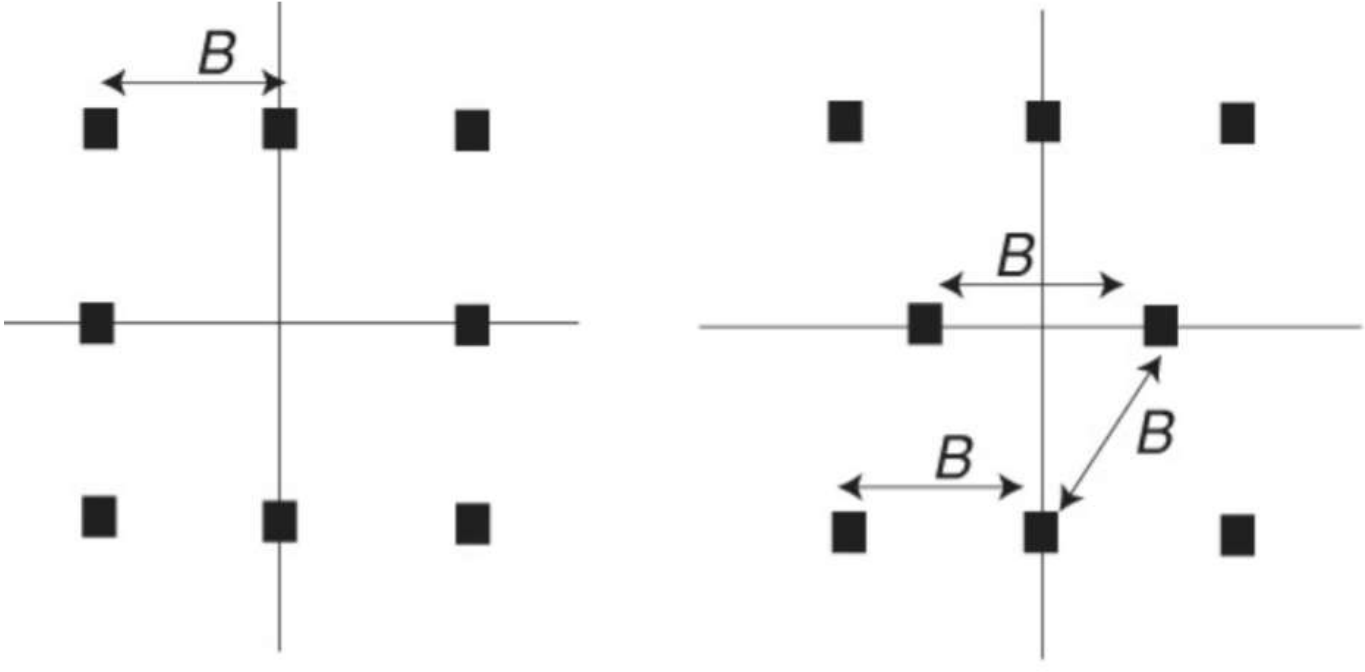
$$E_{\mathrm{s}} = \mathbb{E}_s \left[|s[n]|^2 \right]$$

- Because the constellation is equally likely and zero-mean

$$E_{\mathrm{s}} = \mathbb{E}_s \left[|s[n]|^2 \right] = \frac{1}{M} \sum_{m=0}^{M-1} |s_m|^2$$

- We always normalize constellation energy to control transmit power

Figure 4.4 contains two different 8-point signal constellations. The minimum distance between adjacent points is B. Assuming that the signal points are equally likely in each constellation, find E_s for each constellation and suggest a suitable choice of α to make the constellation have unit energy.



Example

- Perform the operations to obtain a zero mean normalized constellation from

$$\mathcal{C} = \{\mathrm{j}, (1-\mathrm{j})/\sqrt{2}, -\mathrm{j}, (-1-\mathrm{j})/\sqrt{2}\}$$

Constellations

Learning objectives:

- Draw a constellation diagram including bit labels
- Calculate the energy of a constellation and normalize it

Computing the bandwidth, energy, and power of $x(t)$

Learning objectives:

- Calculate bandwidth, energy, and power of a general complex pulse amplitude modulated signal

Bandwidth of a complex pulse amplitude modulated signal

- Complex baseband signal

$$x(t) = \sqrt{E_{\mathrm{x}}} \sum_{n=-\infty}^{\infty} s[n] g_{\mathrm{tx}}(t - nT)$$

- $x(t)$ is a cyclostationary random process
- We can define the power spectral density of signals like this as

$$P_x(f) = \frac{E_x}{T} \mathbb{E}\left[|s[n]|^2\right] |G_{tx}(f)|^2$$

  - With a unit energy constellation

$$P_x(f) = \frac{E_x}{T} |G_{tx}(f)|^2$$

  - $|G_{tx}(f)|^2$ determines the bandwidth of $x(t)$

Energy and Power

- We always assume $\int_{-\infty}^{\infty} |G_{tx}(f)|^2 df = 1$
- Only want to add any energy to the signal through E_x
- The transmit energy is in Joules $\int_{-\infty}^{\infty} P_x(f) df = E_x$
- The transmit power is in Watts E_x/T

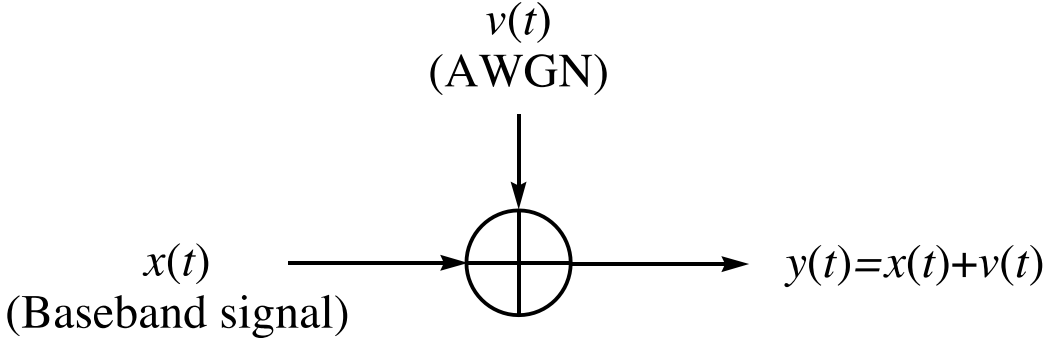
Computing the bandwidth, energy, and power of $x(t)$

Learning objectives:

- Calculate bandwidth, energy, and power of a general complex pulse amplitude modulated signal

Additive white Gaussian noise (AWGN) channel (1)

- Thermal noise is present in any wireless communication system
- AWGN is a good model for the impairments due to thermal noise



- $x(t)$ is the complex baseband signal, $v(t)$ is the AWGN, $y(t)$ is the observation

Additive white Gaussian noise (AWGN) channel (2)

- AWGN means
  1) Noise is additive
  2) $v(t)$ is an IID complex Gaussian random variable

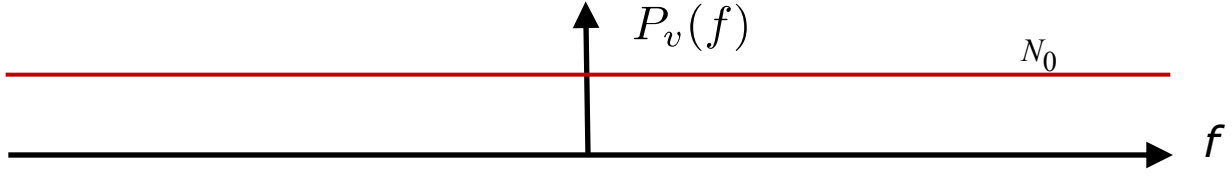
$$f_v(x) = \frac{1}{\pi\sigma^2} e^{\frac{|x-m|^2}{\sigma^2}}$$

- $v(t) \sim \mathcal{N}_c\left(0, \sigma^2\right)$
- $\mathrm{Re}\{v(t)\} \sim \mathcal{N}\left(0, \frac{\sigma^2}{2}\right)$
- $\mathrm{Im}\{v(t)\} \sim \mathcal{N}\left(0, \frac{\sigma^2}{2}\right)$
- $\mathbb{E}\left[\mathrm{Re}\{v(t)\} \cdot \mathrm{Im}\{v(t)\}\right] = 0$

Additive white Gaussian noise (AWGN) channel (3)

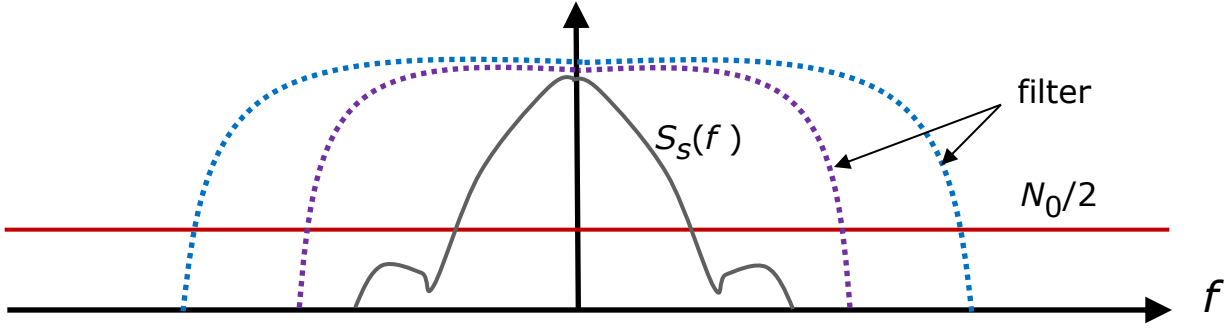
- AWGN means
  - 3) The noise variance is $\sigma_v^2 = N_o = kT_e$
    - k is Boltzman's constant and $k = 1.38 \times 10^{-23} J/K$
    - the effective noise temperature of the device is T_e in Kelvins.
    - Assume $T_e = 290K$ in the absence of other information
  - 4) The noise is White because it is IID

$$R_v(\tau) = \sigma^2(\tau)\delta(\tau) \quad \Longleftrightarrow \quad P_v(f) = \sigma^2$$



Additive white Gaussian noise (AWGN) channel (4)

- Although AWGN is IID in theory, noise PSD in a communication system is limited by the filters in the analog front-end

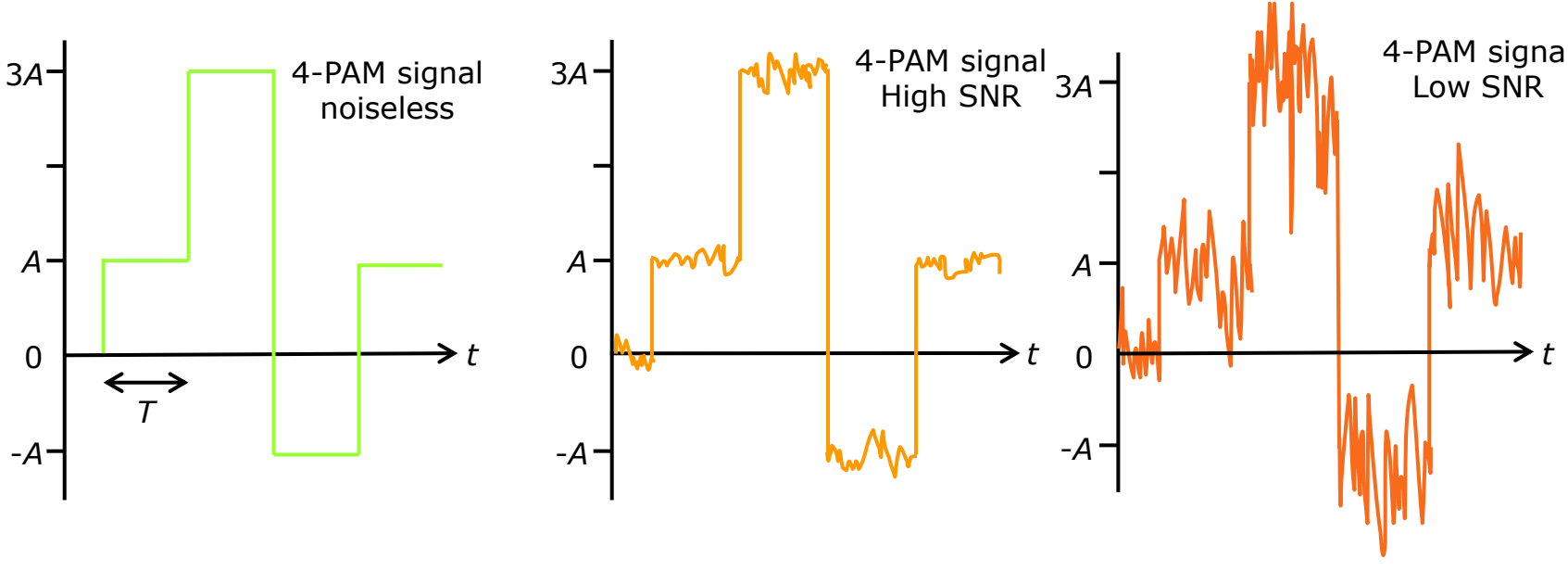


Signal-to-noise ratio (SNR)

- Main performance measure

$$\mathrm{SNR} = \frac{\mathbb{E}_x\left[|x(t)|^2\right]}{\mathbb{E}_v\left[|v(t)|^2\right]}$$

Why it is SNR comes from detection theory, under other assumptions a different conclusion would result



Example

- Consider a wireless communication system where
  - A signal is received with a power of 1 microwatt
  - AWGN at the receiver front-end corrupts the signal
  - Signaling bandwidth is 9MHz
- What is the SNR at the receiver front-end?

Noise power is calculated using:

$$P_n = N_0 \cdot W$$

Where:

- N_0 is the noise power spectral density (W/Hz)
- W is the bandwidth (Hz)

Assuming thermal noise at room temperature:

$$N_0 = kT \approx 4 \times 10^{-21}\,\mathrm{W/Hz}$$

$$W = 9\,\mathrm{MHz} = 9 \times 10^{6}\,\mathrm{Hz}$$

$$P_n = 4 \times 10^{-21} \cdot 9 \times 10^{6} = 3.6 \times 10^{-14}\,\mathrm{W}$$

Additive white Gaussian noise (AWGN) channel

Learning objectives:

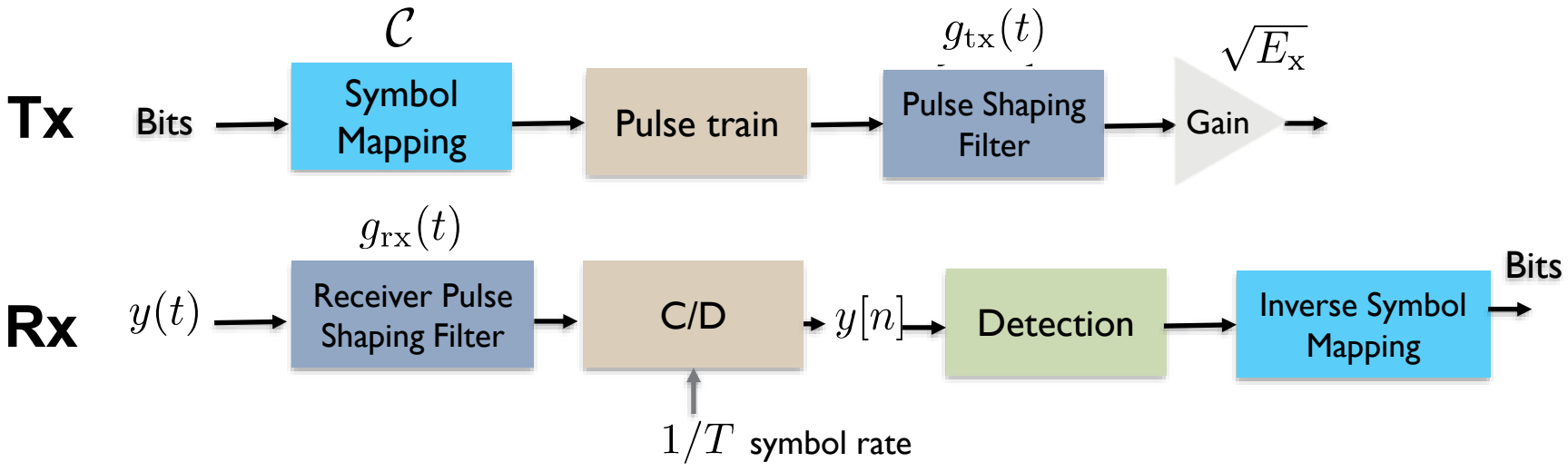
- Define AWGN including the distribution and properties

Overview of AWGN transmitter and receiver

Learning objectives:

- Describe key components of the transmitter and receiver

Overview of AWGN transmitter and receiver



- Problem: how to design $g_{\text{tx}}(t)$ and $g_{\text{rx}}(t)$ to maximize *SINR*

$$\text{SINR} = \frac{\text{S}}{\text{I}+\text{N}}$$

signal power

intersymbol interference

noise power

Pulse shape design

Learning objectives:

- Determine properties of the optimal receiver pulse-shape for the AWGN channel

Pulse shape design

- The $g_{\text{rx}}(t)$ that maximizes SNR is optimal
- Consider the received signal

$$y(t) = \sqrt{E_{\text{x}}} g_{\text{rx}}(t) \star g_{\text{tx}}(t) \star \sum_m s[m]\delta(t - mT) \; + \; g_{\text{rx}}(t) \star v(t)$$

- Let $g(t) = g_{\text{rx}}(t) \star g_{\text{tx}}(t)$
- After sampling

$$y[n] = \sqrt{E_{\text{x}}} \sum_m s[m] g((n-m)T) \; + \; \tilde{v}[n]$$

$m=n$, **desired component**

$m \neq n$, **intersymbol interference**

Signal-to-interference-plus-noise ratio

- The quality of the receiver can be measured by the signal-to-interference-plus-noise ratio (SINR)
  - Signal energy $\mathbb{E}\left|\sqrt{E_{\mathrm{x}}}s[n]g(0)\right|^2 = E_{\mathrm{x}}|g(0)|^2$
  - Noise energy $\mathbb{E}\left|g_{\mathrm{rx}}(t) \star v(t)|_{nT}\right|^2 = N_{\mathrm{o}}\int |G_{\mathrm{rx}}(f)|^2 \mathrm{d}f$
  - Intersymbol interference energy

$$\mathbb{E}\left|\sqrt{E_{\mathrm{x}}}\sum_{m,m\neq n} s[m]g((n-m)T)\right|^2 = E_{\mathrm{x}}\sum_{m\neq 0}|g(mT)|^2$$

  - $$SINR = \frac{E_{\mathrm{x}}|g(0)|^2}{N_{\mathrm{o}}\int |G_{\mathrm{rx}}(f)|^2\mathrm{d}f \; + \; E_{\mathrm{x}}\sum_{m\neq n}|g(mT)|^2}$$

A good receiver maximizes the SINR

SINR maximizing design (1)

- Use series of inequalities to find the SINR maximizing design

$$\frac{E_x|g(0)|^2}{N_o\int|G_{\text{rx}}(f)|^2df\ +\ E_x\sum_{m\neq n}|g(mT)|^2} \leq \frac{E_x\int|G_{\text{rx}}(f)|^2df}{N_o\int|G_{\text{rx}}(f)|^2df\ +\ E_x\sum_{m\neq n}|g(mT)|^2}$$

$$\leq \frac{E_x\int|g_{\text{tx}}(t)|^2dt}{N_o\int|G_{\text{rx}}(f)|^2df}$$

$$= \frac{E_x}{N_o}$$

SINR maximizing design (2)

- Using the Cauchy-Schwarz inequality

$$\begin{aligned} |g(0)|^2 &= \left| \int g_{\text{rx}}(-t) g_{\text{tx}}(t) dt \right|^2 \\ &\leq \int |g_{\text{tx}}(t)|^2 \, dt \int |g_{\text{rx}}(t)|^2 \, dt \\ &= \int |g_{\text{rx}}(t)|^2 \, dt \\ &= \int |G_{\text{rx}}(f)|^2 \, df \end{aligned}$$

- Equality holds when $g_{\text{rx}}(t) = g_{\text{tx}}^*(-t)$
- Matched filter

$$g_{\text{rx}}(t) = g_{\text{tx}}^*(-t)$$

SINR maximizing design (3)

- Zero ISI condition

$$E_x \sum_{m \neq 0} |g(mT)|^2 = 0 \quad \Longleftrightarrow \quad g(nT) = c\delta[n]$$

with our normalization of $g_{\text{tx}}(t)$

$$c = g(0) = \int |g_{\text{tx}}(t)|^2 dt = 1$$

- **Do such pulses exist?**

Pulse shape design

Learning objectives:

- Determine properties of the optimal receiver pulse-shape for the AWGN channel

Nyquist Pulse Shapes

Learning objectives:

- Define Nyquist pulse criterion and determine if a given pulse shape satisfies that criterion.
- Describe key components of the transmitter and receiver

Nyquist pulse-shapes (1)

- Zero ISI condition

$$g(nT) = \delta[n] \quad \Longleftrightarrow \quad \sum_k G\left(f + \frac{k}{T}\right) = T$$

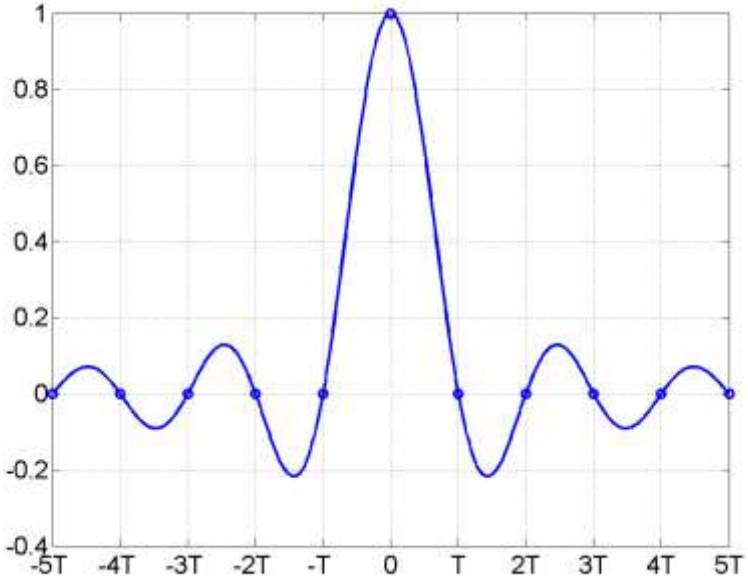
  - or equivalently with $g_d[n] \leftrightarrow G_d(e^{j2\pi f})$

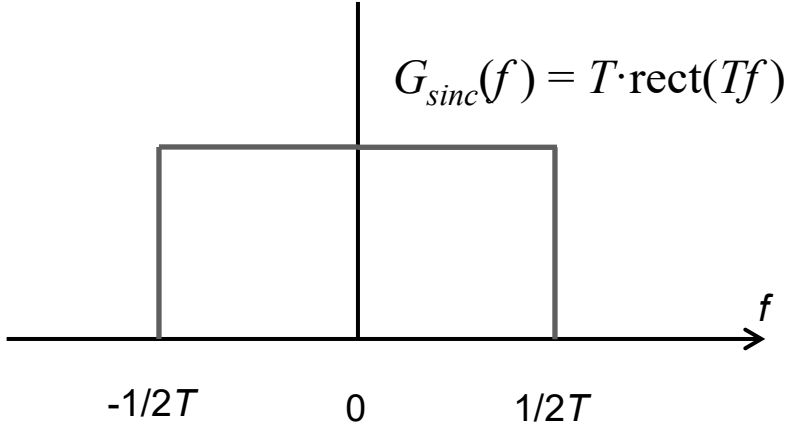
$$G_d(e^{j2\pi f}) = \sum_k G(fT + k) = 1$$

Nyquist pulse shapes (2)

- Example: sinc function

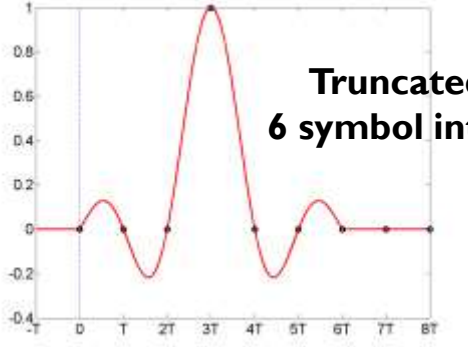
$$g_{\text{sync}}(t) = \frac{\sin(\pi t/T)}{\pi t/T}$$

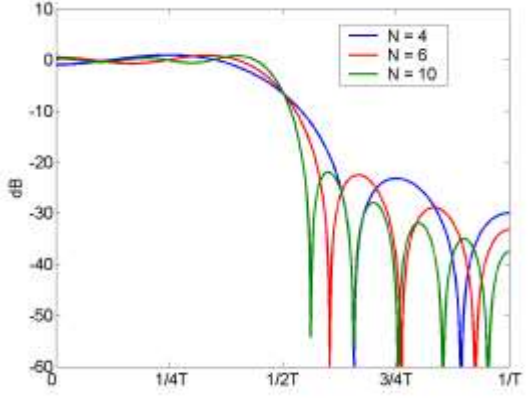




Nyquist pulse shapes (3)

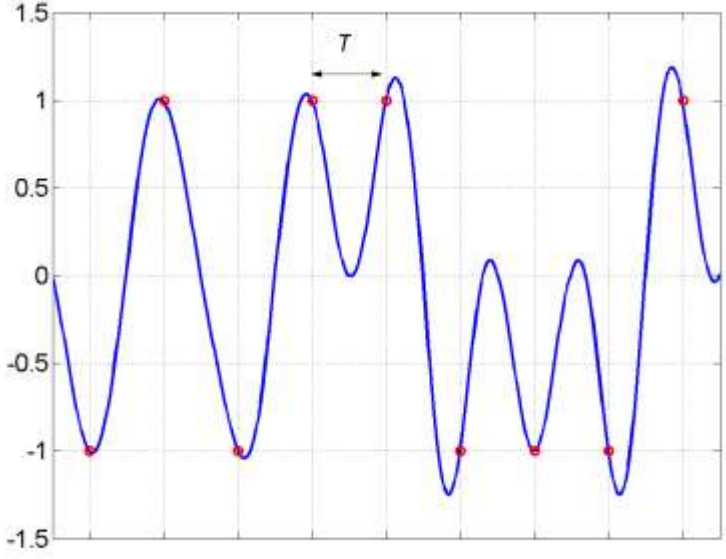
- The sinc pulse shaping filter has a number of problems in practice
  - Must be truncated in practice
  - Is non-causal, but can be made causal through sufficient delay
  - Difficult to approximate steep band edges
  - Wide main pulse implies lots of sensitivity to timing errors
  - Sow decay ~1/t
- When $1/T$ is less than the Nyquist rate we get aliasing!





Nyquist pulse shapes (4)

- BPSK signal using a sinc pulse shaping filter
  - Pulse has been truncated to 20 symbol periods

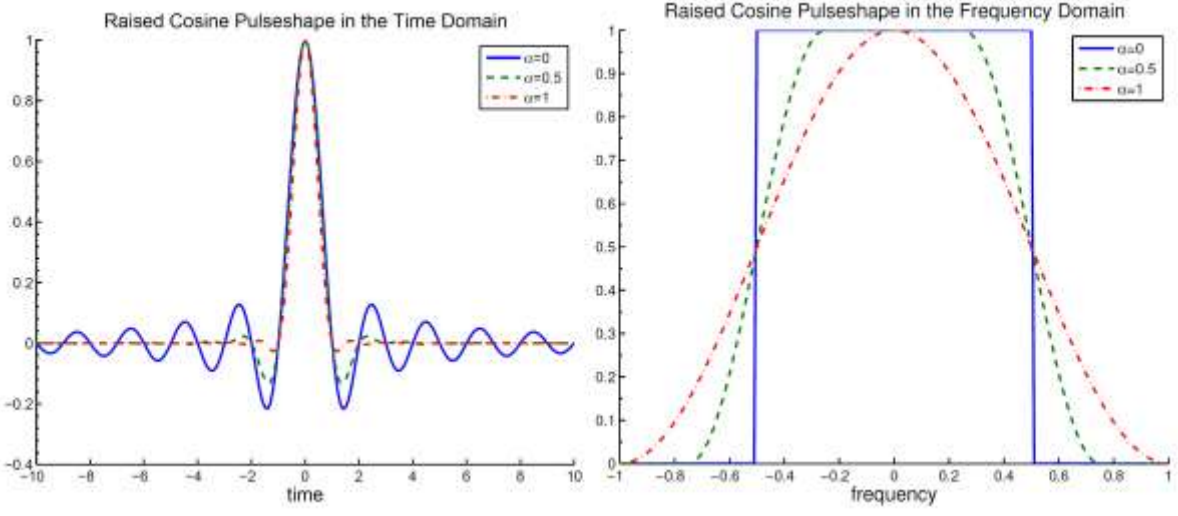


  - Zero ISI

Raised cosine pulse-shape (1)

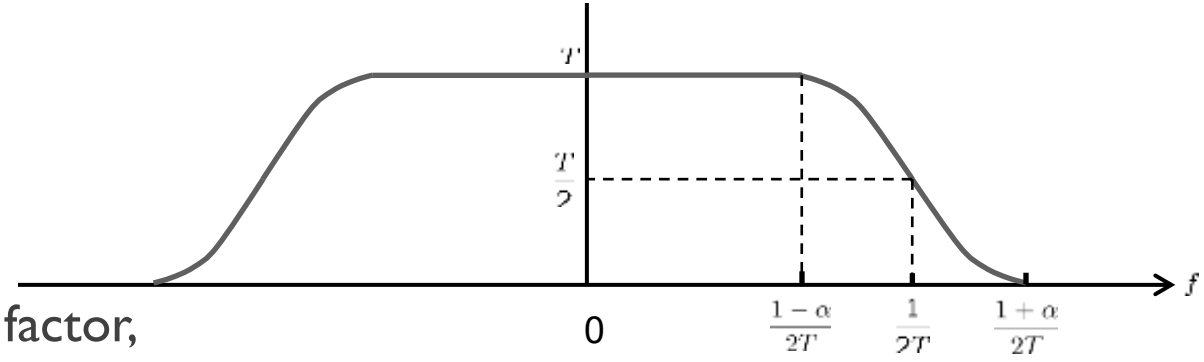
$$G_{\mathrm{rc}}(f) = \begin{cases} T, & 0 \le |f| \le (1-\alpha)/2T \\ \frac{T}{2}\left[1+\cos\frac{\pi T}{\alpha}\left(|f|-\frac{1-\alpha}{2T}\right)\right], & \frac{1-\alpha}{2T} \le \frac{1+\alpha}{2T} \\ 0, & |f| > \frac{1+\alpha}{2T} \end{cases}$$

$$g_{\mathrm{rc}} = \frac{\sin \pi t/T}{\pi t/T}\frac{\cos(\pi\alpha t)}{1-4\alpha^2 t^2/T^2}$$



Raised cosine pulse-shape (2)

- Features



  - α is the rolloff factor,
    - Often expressed as percentage excess bandwidth.
  - $(1+\alpha)/2T$ is the absolute bandwidth of the raised cosine pulse
  - $(1+\alpha)/2T$ is the Nyquist rate of $x(t)$ with a raised cosine pulse shape
  - $1/T$ is the rate at which $x(t)$ is sampled
  - $\alpha = 0$ is the sinc pulse
  - Width of main lobe controlled by α
  - Decay at best ~ $1/t^3$

Square-root raised cosine

- Recall $G(f) = G_{\text{tx}}(f)G_{\text{rx}}(f)$ and for matched filter $G_{\text{rx}}(f) = G_{\text{tx}}^*(f)$
- Use the square-root raised cosine in practice for $g_{\text{tx}}(t)$ and $g_{\text{rx}}(t)$

$$g_{\text{sqrc}}(t) = \frac{4\alpha}{\pi\sqrt{T}} \frac{\cos\left((1+\alpha)\pi t/T\right) + \frac{\sin((1-\alpha)\pi t/T)}{4\alpha t/T}}{1-(4\alpha t/T)^2}$$

Nyquist Pulse Shapes

Learning objectives:

- Define Nyquist pulse criterion and determine if a given pulse shape satisfies that criterion.
- Describe key components of the transmitter and receiver